Technical Presentation

DL3

Computer vision 270302 (slot c)

Name : [illegible]

Reg. No : [illegible]

Q1 Compare different image acquisition devices such as digital cameras, scanners, satellite sensors, and medical imaging systems by explaining their working principles and applications.

Technical Content Accuracy (30)	Problem Identification (25)	use of Tools (20)	Communication (15)	Proffessionalism (10)
25	23	20	15	10

Total (100)
93

Calculator Application – Expression Processing Using Stack

Presented by: Your Name
Department
College

Introduction

- • Calculator evaluates arithmetic expressions.
- • Supports +, −, × , ÷ , %, and ().
- • Follows operator precedence.
- • Produces accurate results quickly.
- • Used in scientific and basic calculators.

Problem Statement

- • Expressions contain multiple operators.
- • Parentheses increase complexity.
- • Correct precedence must be maintained.
- • Efficient evaluation is required.
- • Stack solves these problems.

Recommended Data Structure

- Stack (LIFO)
- • Push and Pop operations.
- • Stores operators temporarily.
- • Handles nested parentheses.
- • Maintains operator precedence.
- • Simple and efficient.

Working Process

- User Input → Infix Expression → Stack → Postfix Conversion → Postfix Evaluation → Final Result

- Example:
- 5 + (3 × 2)
- Postfix: 5 3 2 × +
- Result: 11

Algorithm

- 1. Read expression.
- 2. Push operators to stack.
- 3. Compare precedence.
- 4. Pop higher precedence operators.
- 5. Evaluate postfix.
- 6. Display result.

Efficiency

- • Push: O(1)
- • Pop: O(1)
- • Expression Evaluation: O(n)
- • Memory efficient.
- • Suitable for real-time calculations.

Advantages

- • Fast processing
- • Easy implementation
- • Reduces errors
- • Handles complex expressions
- • Used in compilers and interpreters

Applications

- • Scientific Calculators
- • Mobile Calculator Apps
- • Compiler Design
- • Expression Evaluation
- • Programming Languages

Conclusion

- • Stack is the best data structure for expression processing.
- • Ensures correct evaluation.
- • Improves performance and reliability.

- Thank You!